Lesson Learned from roaster in 2025.

Warming up roaster

1. Heat up the roaster for 20 min. This will allow 1st roast of the day closer to subsequent roasts of the day. Set SV = 20C + desired CT. (look at log book for each bean CT value)
2. 5 min before the actual Charge, reduce the SV to desired CT plus 5C.
3. 5 min before set the SV = CT.

Repeat step 2 and 3 for in between batches. Do this while emptying the chaff tray and getting taking notes.

Green Bean weights

Weight of green beans. The Kaleidos M1 is good between 150-160 g. Typically you will use just 150g batch, but sometime at the end of the bag you have some extras that you can distribute equally to remaining batches. The 2 lb or 5 lb orders work perfect for 150g batch.

Use of Alarms for automation.

This is very easy to setup. Here is an example of various alarms I setup. Once you setup your alarm, this is saved in your alog file, and you don't have to do much to setup the next one. Just adjust the time and/or the set points for different profile.

Alarms

Alarm Table | Alarm Sets

Nr	Status	If Alarm	But Not	From	Time	Source	Condition	Value	Action	Beep	
1	ON	0	0	START	00:01		=	0.0	SV		215
2	ON	0	0	CHARGE	00:01		=	0.0	Slider Burr		99.9
3	ON	0	0	CHARGE	00:03		=	0.0	Slider Air		5
4	ON	0	0	CHARGE	00:50		=	0.0	Slider Burr		90.0
5	ON	0	0	CHARGE	00:53		=	0.0	Slider Air		10
6	ON	0	0	CHARGE	01:35		=	0.0	Slider Burr		80.0
7	ON	0	0	CHARGE	01:38		=	0.0	Slider Air		15

Alarms page from Artisan

The 1st line is an example using EVENTS to trigger an action. This one is to set SV to 215, 1 seconds after the START button is pressed. The Description column is sometimes use for set points.

The 2nd line is an example of setting the burner to 99.9%, 1 second after the CHARGE button is pressed.

The 3rd line is an example of setting the air to 5%, 3 seconds after the CHARGE button is pressed.

Essentially you can set amount of elapsed time after an event to trigger a change in burner or air settings.

The checked mark on the 2nd column (STATUS) can be used to disable without deleting the alarm.

More examples

28	☑	ON	0	0	TP	00:00	BT	>	140.0	Pop Up	☑	check bean color
29	☐	ON	0	0	TP	00:00	BT	>	150.0	DRY	☐	Enter description
30	☑	ON	0	0	TP	00:00	BT	>	120.0	Pop Up	☐	check that SV is 215
31	☑	ON	0	0	DRY END	00:00	ΔBT	<	4.0	Pop Up	☐	BT ROR is too low, watch for stalling, inrease burner power manually
32	☑	ON	31	0	DRY END	00:00	ΔBT	<	4.0	Slider Air	☐	5.0
33	☑	ON	0	0	DRY END	00:00	ΔBT	>	20.0	Pop Up	☐	BT ROR is too high, reduce burner power manually
34	☐	ON	33	0	DRY END	00:00	ΔBT	>	20.0	Slider Air	☐	99.9
35	☐	ON	0	0	DRY END	00:01		>	0.0	Slider Air	☐	99.9
36	☑	ON	0	0	CHARGE	00:01		>	0.0	Slider Drur	☐	99.9
37	☐	ON	0	0	CHARGE	00:01		>	0.0	Slider Air	☐	20.00
38	☑	ON	0	0	DROP	00:01		>	0.0	SV	☐	160.00
39	☑	ON	0	0	DROP	00:01		>	0.0	PID ON	☐	Enter description

More Alarm Examples

You can also use the temperature for example, BT, or ROR (delta BT) to trigger an action. You can trigger a popup, or set Air or Burner to certain values.

Alarms can be saved, and loaded. I have been saving them specific to the bean types for large batches to be able to repeat. Not sure if it will be better later to store this based on processing method vs variety type for the same batch size.

Roast Background

TURN OFF or DO NOT LOAD YOUR BACKGROUND PROFILE AS I THINK IT MESSED UP ARTISAN AUTOMATION FROM THE ALARM.

Roast Control

This section help you setup your burner control.

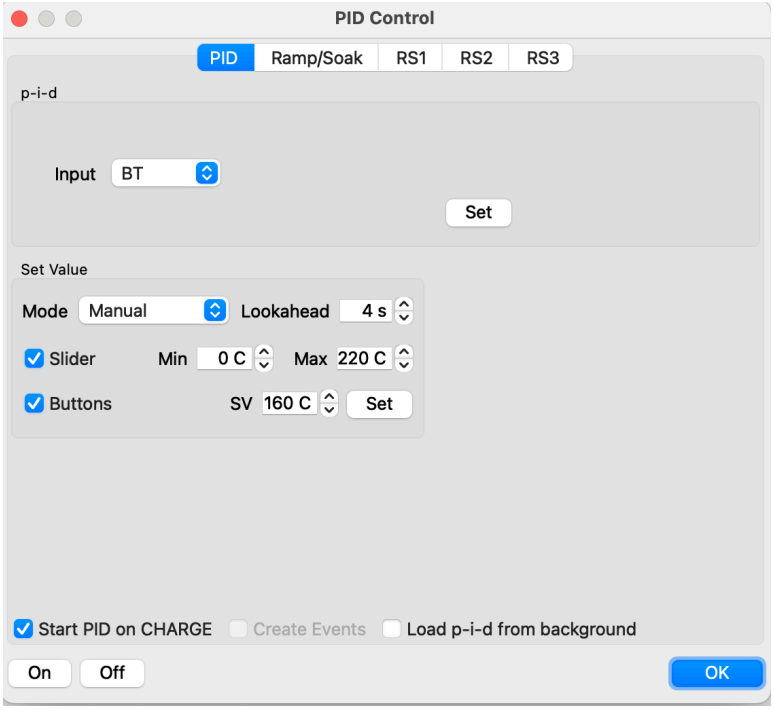
MAKE SURE :

1. The PID is turned on when you click the Auto-Burner as shown below. This is useful when warming up the roaster as the PID will be used to control the burner automatically to the SV setting. Make sure the START HEATING button is pressed for this to commence.

2. The SV is set to CT about typically between 135-145 Celsius for M1, 150-160 grams batch. Hit the set Button if you don't see the SV changed in the graph window.

The SV is automatically set to 215 via the Alarms setup(-automatically). You don't have to do anything after you ensure the alarms is used and enabled.

3. The SV Mode should be left to manual so the burner setting does not change other than via the Alarms.



Caption

4. The Max temperature is set higher than the SV setting in the alarm line 1. So I leave this to 220C.

5. Input is set to BT (Bean Temperature) Probe.

6. Check Start PID on Charge box, this turn on PID automatically when Charge is pressed. This does not do much if the SV is set very high and your current BT is not even close to it.

Other notes, not important when you are roasting based on reducing burner power using Alarms and SV is set way higher than current BT. If the SV is set very close to the current BT, then the burner will be reduced automatically to maintain the temperature at SV setting. When PID is off, the burner is not controlled automatically by Artisan.

Mode = background, this allow you to follow a background profile, Artisan will modulate the burner to change the SV value to match the temperature profile of the background. I am not sure what Load p-i-d from background does but I think it is loading the pid parameters from the background profile. If you never change these parameters, the values from the background profile will be the same with what you are currently using. The on-off button I think just turn on and off the PID control. PID control on = auto-burner, PID control off = Manual burner.

Caption

About Charge Temperature (CT)

Think of this setting as lowering or raising the entire roast curve up and down. If the roast is getting too hot quickly, you should lower your CT. I found that lowering the CT by 5 degrees C is sufficient.

Sometime the batch size also impact CT. The more beans you put in, the slower it will take to roast, so you may have to increase the CT by 5 degrees for every 10g additional above 150g you put in.

Look at your previous profile and adjust accordingly to move up or down on the curve. Remember lowering CT will slow the roast, increasing CT will do the opposite.

START ROASTING

Once everything is setup per the above and you have warm up your Kaleido, you can start roasting.

1. Click ON Button to start recording.
2. Click START Button to start roasting.
3. Load the beans and then open the chute, and
4. …get ready to hit the CHARGE button as the bean drop into the drum.

Your drum and air are setup automatically if you use the alarm setup. Drum speed is set to 100%, I never change this. I think keeping drum speed high prevent pitting, I never know why you don't want this to be low at all in a roaster this small. Air start slow and gradually increase. My idea is to keep the heat as much as possible until later to control the ROR and suck the chaff out the drum.

About Development Time Ratio (%DTR vs absolute time after FC)

Keep this under the following rule of thumb, and take it with a grain of salt. I learned that the % DTR is not very useful, or less useful compare to using color and weight loss. Probably better in the future to improve this note to use development time measured after FC. So instead of percentage of total time, just indicate 1 min after or 1m:30s after FC (this is what I refer as Absolute time after FC method).
The notes below on DTR is from when I was still using it for guidance, I don't use them much at all anymore.

Light to Dark Scale

- **Light Roast:** 15% – 20% DTR. Often dropped mid-crack or immediately after first crack ends to preserve high acidity and origin character.
- **Medium Roast:** 20% – 25% DTR. The "sweet spot" for most specialty roasters to balance acidity with caramelization and body.
- **Dark Roast:** 25% – 30%+ DTR. Roasts typically pushed toward or into the second crack, prioritizing smoky, bittersweet, and chocolate notes over acidity.

City to Full City Scale
This scale is more granular and based on visual markers alongside DTR:

- **City:** 12% – 16% DTR. Dropped early in the first crack; results in high acidity and potentially bready or grassy notes if not fully developed.
- **City Plus (City+):** 16% – 20% DTR. Dropped just as first crack concludes; the "crossroads" between light and medium with balanced acidity.
- **Full City:** 20% – 25% DTR. Roasted until the beans are smooth and just before the first signs of the second crack; standard medium roast.
- **Full City Plus (Full City+):** 25% – 30%. Dropped at the first few audible snaps of the second crack; a dark roast with some surface oil.

End of Roast (EOR)

When beans have cooled down, you can end the roast and the save option will show up. Name the file with the bean name, followed with the batch number, and that's it. Later you can go back after you measure the bean weight lost to enter them in the file name.

Roast Properties

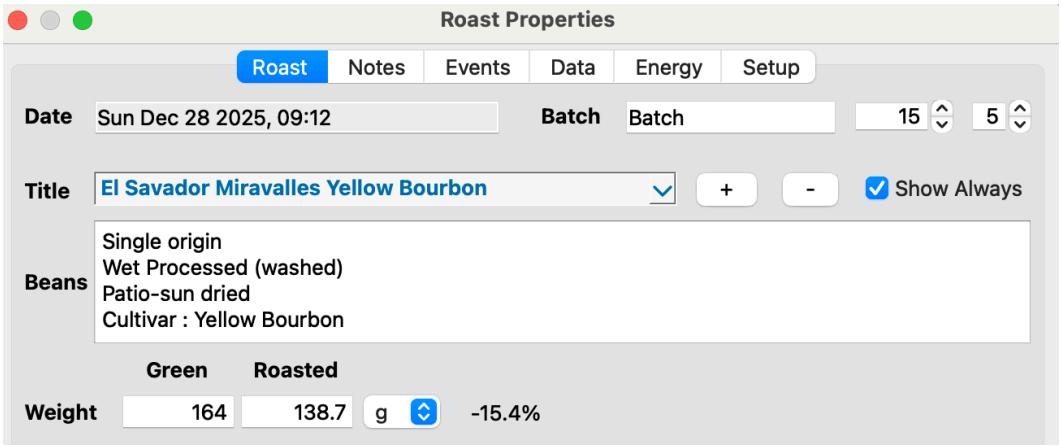
The Title will show on the curve page. Keep it short, I just use the bean name from Sweet Maria.

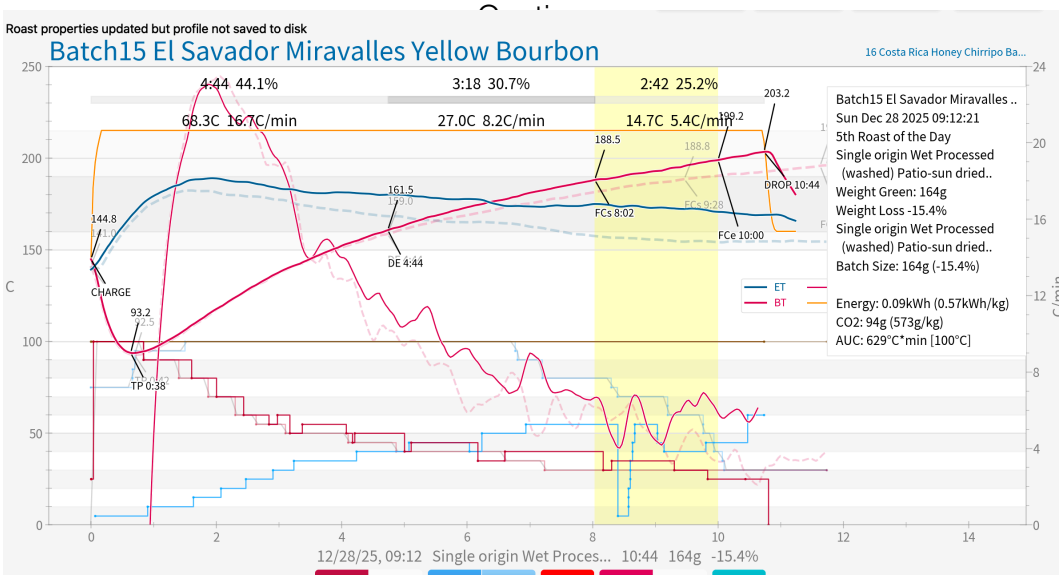
The Beans section, you can keep more detail here like the processing method, variety, etc.

The Batch field, write the word Batch, it shows up preceding the Title.
The number 15 after it will increment automatically to 16 at the subsequent roast but you can override it. The smaller number after that (5) reflects the 5th roast of the day.

Weight, if you enter them, it will be displayed and auto-calculate weight lost. A nice thing to have.

If you have different bean you are roasting during the day, change the title, and beans field accordingly. You can also open previously used artisan log (*.alog) file and reused a lot of the filled out information from previous roast.





Caption